

# FIDELITY IS NOT UTILITY: A CONTROLLED DISSOCIATION IN LEARNED IMAGE TRANSLATION

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## ABSTRACT

Image translation for harmonization and domain adaptation is routinely selected on distributional fidelity metrics such as the Fréchet Inception Distance (FID), on the assumption that a more target-like image is more useful: a team can lower FID and ship a model that is worse at the task. That fidelity can mis-rank generators is known; we make a sharper, mechanistic claim: the utility loss is a property of the learned distribution-matching *objective* itself, not of any architecture, which is why it survives the move from GANs to diffusion. Our instrument is a *frozen* downstream evaluator scored alongside FID on identical inputs, plus a matched non-learned control. On GTA5→Cityscapes ( $N=1000$ , frozen SegFormer), CycleGAN more than halves FID ( $137.2 \rightarrow 65.5$ ) while mIoU falls  $0.380 \rightarrow 0.167$ ; the non-learned control preserves mIoU (0.363) without improving FID. Sweeping SDEdit’s noise strength traces a monotonic frontier: FID  $145.1 \rightarrow 88.0$  as mIoU falls  $0.323 \rightarrow 0.106$ , with seed standard deviation at most 0.005 mIoU. Two controls close the obvious escapes: with an empty prompt, fidelity is worse than the untranslated source (FID 154.2) yet mIoU still collapses (0.170), decoupling the damage from any fidelity gain; and edge-conditioned ControlNet does not escape the frontier (mIoU 0.221 at FID 133.2, versus 0.289 at matched FID). Multi-site brain-tumor MRI reproduces it: frozen nnU-Net tumor Dice falls  $0.682 \rightarrow 0.064$  under learned harmonization but holds at 0.669 non-learned. Across both generative families and both domains, the low-FID/high-utility corner stayed empty. We do not claim fidelity is uninformative, only that it is non-monotone in utility once a learned prior enters the loop; our contribution is a measurement and a protocol, not a method.

## 1 INTRODUCTION

Distribution-matching translation is trained and evaluated to make outputs *look like* a target domain. The implicit promise is that looking more like the target makes outputs more *useful*—for example, that harmonizing site  $B$  to site  $A$  helps a model trained on  $A$ . We test that promise directly and find it false for learned translation.

Our contributions are:

- A **controlled protocol** for separating fidelity from utility: a frozen downstream evaluator plus a matched non-learned control, measuring FID and the task metric on identical inputs (Section 3).
- Evidence that the fidelity–utility dissociation is a property of the **learned distribution-matching objective**, not a single architecture—it reproduces under both a GAN (CycleGAN) and a diffusion model (SDEdit).
- A **continuous characterization** via SDEdit strength: a monotonic FID-vs-mIoU frontier with tight seed variance, an empty-prompt ablation that decouples utility loss from fidelity gain, and a structure-preserving (ControlNet) baseline that fails to escape the frontier.
- **Cross-domain** generality: the same dissociation in driving scenes and in clinical brain-tumor MRI.

## 2 RELATED WORK

Three separately-established observations converge in our claim; we unify them.

**Fidelity metrics mis-rank generators for downstream tasks.** Konz et al. (2024) show no perceptual metric reliably tracks segmentation quality in medical translation; Dutt et al. (2025) show low FID does not predict clinical utility; and Kynkäänniemi et al. (2023) show FID is gameable by aligning ImageNet-class statistics. These are *metric-evaluation* studies; we add a learned/non-learned control and a second generative family and domain.

**Distribution-matching translation alters task content.** Cohen et al. (2018) show CycleGAN-style matching can hallucinate or remove tumors to fit the target distribution, and semantic-consistency losses (Hoffman et al., 2018; Peng et al., 2023) exist precisely to patch this. We show that *without* such non-distributional anchoring the pure learned objective degrades utility, reframing those losses as evidence for our thesis.

**Non-learned alignment rivals learned translation.** Yang & Soatto (2020) reach strong segmentation adaptation with a non-learned spectral swap. We use histogram/color transfer as the analogous control and, unlike FDA, measure the fidelity axis explicitly to expose the dissociation. Richter et al. (2022) likewise treat a segmentation-stability metric as a guardrail that realism must not break.

**Positioning.** No prior work runs the clean  $2 \times 2$  {learned, non-learned}  $\times$  {fidelity $\downarrow$ ?, utility $\downarrow$ ?} as its thesis, ties the failure to the objective across both a GAN *and* a diffusion model, and does so across driving and medicine. This is a measurement/positioning contribution, not a new architecture.

## 3 SETUP: SEPARATING FIDELITY FROM UTILITY

Given source images  $x$  and a translation  $T$ , we evaluate  $T(x)$  on two axes. **Fidelity** is  $\text{FID}(T(x), \text{target})$  via clean-fid (Parmar et al., 2022); lower is closer to the target. **Utility** is a *frozen* downstream model  $f$  (never retrained) applied to  $T(x)$  and scored against ground truth; because  $f$  is fixed, any change in its score is attributable purely to  $T$ . We compare four families of  $T$ : (i) identity (raw source, reference); (ii) non-learned (histogram / color transfer); (iii) learned-GAN (CycleGAN, Zhu et al., 2017); and (iv) learned-diffusion (SDEdit, Meng et al., 2022, over a sweep of noise strengths, plus ControlNet-Canny, Zhang et al., 2023). FID is compared only at equal sample size, which it is known to bias.

## 4 EXPERIMENTS

### 4.1 DRIVING: GTA5 $\rightarrow$ CITYSCAPES

The source is GTA5 (Richter et al., 2016) and the target is Cityscapes (Cordts et al., 2016). The frozen evaluator is a Cityscapes-finetuned SegFormer-b4 (Xie et al., 2021) (19 classes); utility is mIoU. We use  $N=1000$  GTA5 images per condition, with the Cityscapes validation set (500 images) as the fixed FID reference, and identical evaluator and clean-fid (GPU) for all conditions. SDEdit runs on Stable Diffusion (Rombach et al., 2022) and rows report mean  $\pm$  std over three seeds {42, 43, 44}. Results are in Table 1 and Figure 1.

**Findings.** Figure 2 shows representative translations. (a) Learned translation lowers FID relative to raw while collapsing mIoU by 15–72%; CycleGAN more than halves FID (137 $\rightarrow$ 65) at mIoU 0.167. (b) The non-learned baseline does not improve FID yet preserves mIoU (−1.8%). (c) SDEdit strength traces a **monotonic frontier** with tight seed variance ( $\sigma(\text{mIoU}) \leq 0.005$ ): as strength rises, FID falls (145 $\rightarrow$ 88) and mIoU falls with it (0.323 $\rightarrow$ 0.106); no learned setting reaches the ideal corner, occupied only by raw/non-learned. (d) **Empty-prompt ablation.** With no text prompt, FID is *worse* than raw (154.2  $>$  137.2) yet mIoU still collapses (0.170)—the utility loss is the diffusion prior itself, not text guidance, and occurs even absent any fidelity gain. (e) **Structure-preserving baseline.** Conditioning on the source image’s Canny edges (ControlNet-Canny) does *not* rescue utility: it lands on the learned frontier (FID 133.2, mIoU 0.221 $\pm$ 0.002), Pareto-dominated by

Table 1: GTA5→Cityscapes,  $N=1000$ , frozen SegFormer-b4 + clean-fid. SDEdit rows are mean  $\pm$  std over three seeds. Learned translation lowers FID while collapsing mIoU; the non-learned baseline preserves mIoU without improving FID; no learned setting reaches the low-FID/high-mIoU corner.

| condition                 | family           | FID $\downarrow$ | mIoU $\uparrow$   | $\Delta$ mIoU |
|---------------------------|------------------|------------------|-------------------|---------------|
| raw GTA5                  | baseline         | 137.2            | 0.380             | —             |
| color match               | non-learned      | 165.9            | 0.363             | −0.018        |
| SDEdit 0.20               | diffusion        | 145.1 $\pm$ 1.8  | 0.323 $\pm$ 0.003 | −0.057        |
| SDEdit 0.30               | diffusion        | 133.1 $\pm$ 2.1  | 0.289 $\pm$ 0.002 | −0.092        |
| SDEdit 0.40               | diffusion        | 116.4 $\pm$ 1.1  | 0.237 $\pm$ 0.003 | −0.143        |
| SDEdit 0.50               | diffusion        | 100.3 $\pm$ 1.4  | 0.194 $\pm$ 0.005 | −0.187        |
| SDEdit 0.60               | diffusion        | 90.8 $\pm$ 2.0   | 0.150 $\pm$ 0.001 | −0.231        |
| SDEdit 0.70               | diffusion        | 88.0 $\pm$ 0.4   | 0.106 $\pm$ 0.001 | −0.274        |
| CycleGAN                  | learned-GAN      | 65.5             | 0.167             | −0.214        |
| SDEdit 0.50, empty prompt | diffusion (abl.) | 154.2 $\pm$ 1.9  | 0.170 $\pm$ 0.003 | −0.211        |
| ControlNet-Canny          | diffusion + edge | 133.2 $\pm$ 0.3  | 0.221 $\pm$ 0.002 | −0.159        |

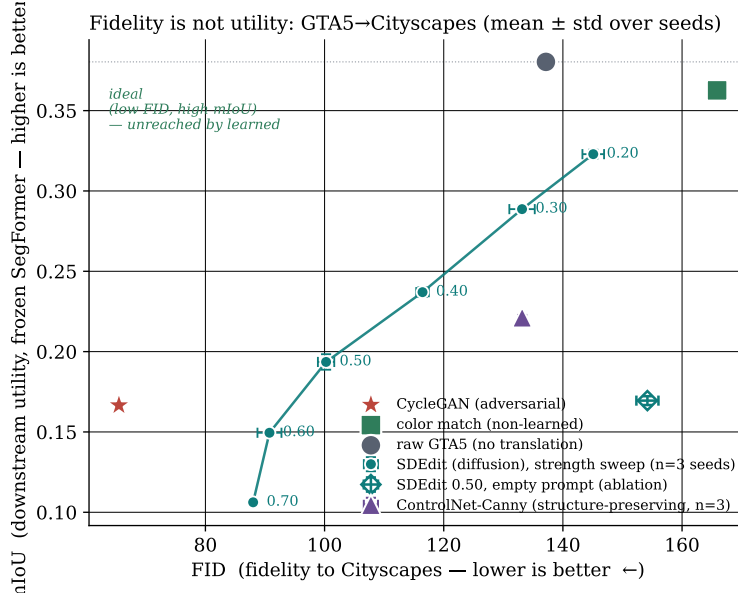


Figure 1: The fidelity–utility plane for GTA5→Cityscapes (mean  $\pm$  std over seeds). The SDEdit strength sweep traces a monotonic frontier: as strength rises, FID falls and mIoU falls with it. Every learned point (CycleGAN, all SDEdit strengths, the empty-prompt ablation, and edge-conditioned ControlNet) sits well below raw and the non-learned baseline on utility; the ideal low-FID/high-mIoU corner is empty.

SDEdit at equal FID (SDEdit 0.30: FID 133.1, mIoU 0.289). Edge structure is not the task-relevant signal; only conditioning on the segmentation *map* (Peng et al., 2023) recovers utility, and only by injecting the labels—the non-distributional constraint our thesis requires. A local  $N=200$  run reproduces the same ordering on an independent machine.

#### 4.2 MEDICAL: MULTI-SITE BRAIN-TUMOR MRI

The frozen evaluator is an nnU-Net (Isensee et al., 2021) trained on BraTS (Menze et al., 2015) and frozen; utility is tumor Dice on the external UPenn-GBM site (Bakas et al., 2022) ( $n=103$ ). The learned harmonizer is SA-CycleGAN-2.5D, a self-attention CycleGAN over 2.5D multi-modal MRI. Table 2 reports the result.



Figure 2: Qualitative GTA5→Cityscapes translations (three scenes, seed 42). Left to right: raw source, non-learned color match, CycleGAN, SDEdit at increasing strength, the empty-prompt SDEdit, and edge-conditioned ControlNet. As the learned translations become more Cityscapes-like they visibly hallucinate and alter scene structure—buildings, lane markings, added objects—that the frozen segmenter depends on, while the non-learned baseline leaves structure intact.

Table 2: External UPenn-GBM tumor Dice under a frozen BraTS-trained nnU-Net. Learned harmonization collapses Dice even though it makes images more source-like; the non-learned baseline preserves it.

| input to frozen nnU-Net           | family                  | Dice (fg. mean) |
|-----------------------------------|-------------------------|-----------------|
| BraTS test (in-domain ref.)       | —                       | 0.771           |
| UPenn raw (external)              | baseline                | 0.682           |
| UPenn + histogram match           | non-learned             | 0.669           |
| UPenn + learned harmonization     | learned-GAN             | 0.064           |
| UPenn + symmetric seg-consistency | learned-GAN (+ content) | 0.205           |

**Findings.** Learned harmonization collapses external-site Dice (0.682→0.064) while the non-learned baseline preserves it (0.669). Content/segmentation-consistency losses reduce but do not eliminate the loss (0.205  $\ll$  0.682), consistent with the driving result that the degradation is intrinsic to the learned objective. A second external site (RHUH-GBM, Cepeda et al., 2023,  $n=40$ ) confirms the direction (0.718→0.599, −16.6%). We note two evaluation protocols exist: the headline numbers use the frozen independent nnU-Net; a separate method-development evaluator ranks the symmetric loss as the best-preserving *learned* harmonizer, which belongs to a companion method paper.

#### 4.3 WHERE THE DAMAGE FALLS

The headline mIoU collapse hides structure. Breaking IoU out per class and grouping the 19 Cityscapes classes into large amorphous regions and thin, small, label-carrying objects (Table 3) shows the degradation is highly *selective*. The non-learned control loses a few percent on both groups, almost uniformly. Every learned translation instead loses far more on the thin classes than on the large ones: SDEdit at strength 0.50 costs 27% of large-region IoU but 83% of thin-object IoU, and CycleGAN 50% against 70%. Poles, traffic lights, traffic signs, riders and motorcycles are precisely the structures a segmenter needs and precisely what the learned prior does not preserve. Edge-conditioned ControlNet sits in between (27% / 63%): conditioning on edges recovers some thin structure, consistent with Section 3’s argument that what is missing is a non-distributional constraint, but not enough to escape the frontier.

Table 3: Per-class IoU grouped into large amorphous regions (road, sidewalk, building, vegetation, terrain, sky) and thin/small label-carrying objects (pole, traffic light, traffic sign, rider, motorcycle); classes whose raw IoU is below 0.05 are excluded, since a ratio off a near-zero baseline is noise. The non-learned control degrades both groups mildly and almost uniformly; every learned translation degrades the thin classes far more than the large ones.

| group                | raw   | color match | CycleGAN     | SDEdit 0.50  | ControlNet   |
|----------------------|-------|-------------|--------------|--------------|--------------|
| large regions        | 0.661 | 0.630 (−5%) | 0.328 (−50%) | 0.481 (−27%) | 0.482 (−27%) |
| thin / small objects | 0.174 | 0.168 (−4%) | 0.052 (−70%) | 0.030 (−83%) | 0.065 (−63%) |

## 5 ANALYSIS

The learned objective optimizes appearance under a distribution/adversarial or denoising target with no term guaranteeing task-relevant signal survives. SDEdit strength is precisely the amount of learned prior injected, and utility falls monotonically with it. The empty-prompt result decouples utility loss from fidelity gain: passing through the prior damages task content even when the output is *less* target-like. Non-learned transforms, lacking a learned prior, cannot introduce this failure; edge-conditioned ControlNet, injecting structure but not semantics, does not avoid it.

## 6 LIMITATIONS

FID is sample-size biased; all conditions use the same  $N$  (1000 translated vs the 500-image Cityscapes validation set as the fixed reference), and SDEdit points report mean  $\pm$  std over three seeds ( $\sigma(\text{mIoU}) \leq 0.005$ ,  $\sigma(\text{FID}) \leq 2.1$ ), so the ordering is robust to both sample size and seed. The structure-preserving baseline uses edge (Canny) conditioning; a segmentation-map-conditioned variant would recover utility only by injecting the labels, which concedes the thesis rather than refuting it.

## 7 CONCLUSION

Fidelity is not utility. Rewarding distributional realism can actively harm the downstream task for learned translation, across architectures and domains. We recommend reporting a frozen-evaluator utility axis alongside any fidelity metric, and treating non-learned baselines as first-class controls.

## REPRODUCIBILITY STATEMENT

All conditions regenerate from a single command; seeds are fixed and logged; frozen evaluators are referenced by checkpoint id (SegFormer-b4, nnU-Net); FID uses clean-fid at a fixed  $N$ . Code, configs, and the  $N=1000$  result set are released with the paper.

## A FULL PER-CLASS IOU

Table 4 reports every Cityscapes class, the breakdown behind Table 3. SDEdit and ControlNet columns are means over the three seeds. Classes with very small raw IoU (*train*, *bicycle*) are near-absent in the GTA5 subset; a relative change computed off such a baseline is noise, which is why the grouped statistics in Table 3 exclude classes with raw IoU below 0.05. The selectivity is visible directly: *road* loses little under every condition, while *traffic light*, *traffic sign*, *person* and *rider* are almost entirely destroyed by the learned translations and almost untouched by the non-learned control.

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Table 4: Per-class IoU under a frozen SegFormer-b4,  $N=1000$ .

| class         | raw   | color match | CycleGAN | SDEdit 0.50 | ControlNet |
|---------------|-------|-------------|----------|-------------|------------|
| road          | 0.937 | 0.932       | 0.843    | 0.845       | 0.884      |
| sidewalk      | 0.540 | 0.502       | 0.202    | 0.280       | 0.309      |
| building      | 0.711 | 0.681       | 0.356    | 0.504       | 0.505      |
| wall          | 0.356 | 0.315       | 0.089    | 0.077       | 0.094      |
| fence         | 0.102 | 0.111       | 0.020    | 0.021       | 0.031      |
| pole          | 0.269 | 0.252       | 0.112    | 0.095       | 0.159      |
| traffic light | 0.073 | 0.067       | 0.008    | 0.008       | 0.012      |
| traffic sign  | 0.069 | 0.063       | 0.012    | 0.017       | 0.022      |
| vegetation    | 0.649 | 0.625       | 0.364    | 0.455       | 0.427      |
| terrain       | 0.227 | 0.159       | 0.083    | 0.050       | 0.011      |
| sky           | 0.901 | 0.883       | 0.121    | 0.750       | 0.755      |
| person        | 0.270 | 0.240       | 0.035    | 0.024       | 0.111      |
| rider         | 0.182 | 0.156       | 0.028    | 0.015       | 0.085      |
| car           | 0.737 | 0.725       | 0.384    | 0.226       | 0.447      |
| truck         | 0.666 | 0.608       | 0.198    | 0.197       | 0.175      |
| bus           | 0.248 | 0.259       | 0.206    | 0.097       | 0.127      |
| train         | 0.002 | 0.002       | 0.000    | 0.001       | 0.001      |
| motorcycle    | 0.279 | 0.303       | 0.101    | 0.016       | 0.046      |
| bicycle       | 0.008 | 0.007       | 0.001    | 0.001       | 0.004      |

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